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Final Project Report

Section: B2 Group: 06

FPGA-Based Visualization of 31-bit Hamming Code for Error Detection and Correction

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1 Abstract

This project implements a 31-bit Hamming code error correction system on a TerasIC DE1-SoC FPGA board using Verilog hardware description language. The system enhances data reliability by detecting and correcting single-bit errors in a 26-bit data input, expanded with 5 parity bits to form a 31-bit Hamming code. Data input is facilitated through the FPGA's built-in slide switches and push buttons, while outputs are displayed on the six 7-segment displays. The design includes modules for data encoding, error detection, and correction, enabling real-time visualization of the entire process. Additionally, a 15-bit Hamming code learning module is developed using a CPLD platform, providing an educational perspective. The integration of both modules highlights the significance of error correction in ensuring data integrity in modern communication systems.

2 Introduction

Error correction plays a crucial role in digital communication, ensuring reliable data transmission by detecting and correcting errors caused by noise or other disruptions. Among various error-correcting codes, Hamming codes are widely used due to their simplicity and effectiveness in handling single-bit errors. This project presents a comprehensive implementation of the 31-bit Hamming code using an FPGA, focusing on real-time error correction and system visualization. To enhance learning, a 15-bit Hamming code module is also developed on a CPLD platform, providing an educational tool for understanding error correction fundamentals. By combining these implementations, the project emphasizes both practical application and theoretical understanding, fostering a deeper appreciation for the importance of robust error-correcting codes in maintaining data integrity across digital systems.

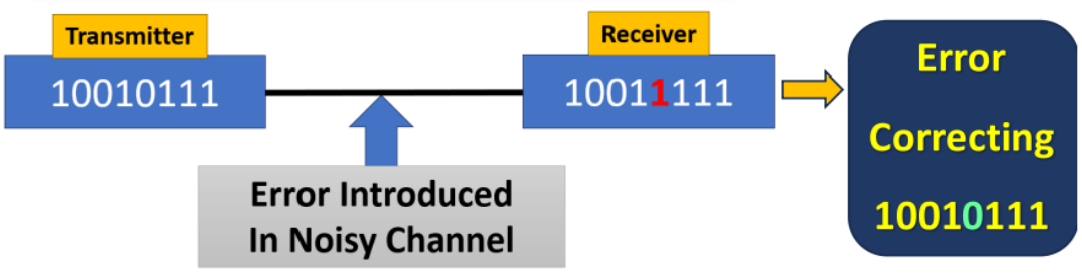


Fig: Simple block diagram of error correcting in digital data transmission.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of the project "FPGA-Based Visualization of 31-bit Hamming Code for Error Detection and Correction" is defined by the objectives of implementing an efficient hardware-based error detection and correction system, leveraging FPGA capabilities, and integrating real-time visualization. The project focuses on the following key areas:

1. Error Detection and Correction System
Implementation of 31-bit Hamming Code: The project encompasses the design of a robust hardware implementation of the 31-bit Hamming code, which is capable of detecting single-bit and correcting single-bit errors in data.
Error Injection and Validation: The system includes provisions to simulate error injection, allowing for thorough testing and validation of the error correction process.

2. FPGA Utilization
Hardware Platform: The project uses an FPGA as the primary hardware platform due to its ability to perform parallel processing and its reconfigurability for diverse applications.
Real-Time Processing: The system will exploit the high-speed computational capabilities of FPGAs to achieve real-time error detection and correction.
Resource Optimization: Efficient utilization of FPGA resources such as logic elements, memory, and interconnects will be a priority to ensure scalability and portability.

3. Visualization Module
User-Friendly Interface: The project will develop a real-time visualization module that displays the step-by-step operation of the 31-bit Hamming code, including encoding, error detection, and correction.

3.1.2 Literature Review

The literature review highlights the foundational concepts and advancements relevant to implementing a 31-bit Hamming code for error detection and correction using FPGA technology. Hamming codes, known for their simplicity and efficiency, have been extensively studied for their ability to detect and correct single-bit errors, making them suitable for digital communication and storage applications. Theoretical research provides robust algorithms for encoding, detecting, and correcting errors, forming the basis for hardware implementations.

Field-Programmable Gate Arrays (FPGAs) are identified as ideal platforms for real-time error correction due to their high-speed processing, parallelism, and reconfigurability. Research demonstrates the importance of efficient hardware resource utilization and the use of hardware description languages like Verilog or VHDL for designing error correction circuits.

The integration of visualization into FPGA-based systems has proven valuable for debugging, educational purposes, and analyzing algorithmic behavior. Real-time visualization of the Hamming code's operation aids in understanding and monitoring the error correction process. Although Hamming codes are limited to single-bit error correction, their simplicity and low-latency operation make them well-suited for applications in embedded systems, memory modules, and communication systems.

This project builds on existing research to develop a robust FPGA-based implementation of 31-bit Hamming code with integrated real-time visualization, aiming to provide both practical functionality and educational utility.

3.1.3 Formulation of Problem

Data integrity in digital systems is vital, especially in error-prone environments. Hamming codes are efficient for detecting and correcting single-bit errors, but implementing and visualizing these algorithms in real-time presents challenges, particularly in high-volume data processing. Field-Programmable Gate Arrays (FPGAs) offer high-speed, parallel processing solutions for such tasks, though implementing the 31-bit Hamming code on FPGA with a visualization mechanism remains complex. This project aims to design an FPGA-based system that performs real-time error detection and correction while integrating a user-friendly visualization to enhance understanding and analysis. Here 26 bit data input is taken and 5 bit is used for parity bits. An error is injected in a random bit position and it will detect and correct the error and return the given data input.

3.1.4 Analysis

Developing a Learning Module for (15,11,4) Hamming Code System
Working Method:
Here, total number of data bits will be 15,
Number of message bits will be 11,
And there will be 4 parity bits.
Let, the 4 parity bits are P1, P2, P4, P8
P1 checks all the bit positions where the LSB is 1 in binary representation.
So, P1 checks those bits position whose binary representation is in the form of (X X X 1). And, P1 stays at the first (0001) bit position.
Similarly,

Parity Bit	Checked Bit's Binary Representation Form	Staying Position
P_2	X X 1 X	2 (0010)
P_4	X 1 X X	4 (0100)
P_8	1 X X X	8 (1000)

Suppose, our 11 message bits are:
D11, D10, D9, D8, D7, D6, D5, D4, D3, D2, D1
So, total transferred bits will be:

D_{11}	D_{10}	D_9	D_8	D_7	D_6	D_5	P_8	D_4	D_3	D_2	P_4	D_1	P_2	P1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

If we place those data bits in 4x4 table, then

	0001^{1} P_1	0010^{2} P_2	0011^{3} D_1
0100^{4} P_4	0101^{5} D_2	0110^{6} D_3	0111^{7} D_4
1000^{8} P_8	1001^{9} D_5	1010^{10} D_6	1011^{11} D_7
1100^{12} D_8	1101^{13} D_9	1110^{14} D_{10}	1111^{15} D_{11}

Every parity bit will check its corresponding message bits and take a value between 1 and 0 so that even number of 1 will be present in every region.

	0001^{1} P_1	0010^{2} P_2	0011^{3} D_1
0100^{4} P_4	0101^{5} D_2	0110^{6} D_3	0111^{7} D_4
1000^{8} P_8	1001^{9} D_5	1010^{10} D_6	1011^{11} D_7
1100^{12} D_8	1101^{13} D_9	1110^{14} D_{10}	1111^{15} D_{11}

There will be even number of '1' in this highlighted region where each data bit will be checked by P1.
So, **P1=D1⊕ D2⊕ D4 ⊕ D5 ⊕ D7⊕ D9⊕ D11**

Similarly,

	$^{1}_{0001}$ P_1	$^{2}_{0010}$ P_2	$^{3}_{0011}$ D_1
$^{4}_{0100}$ P_4	$^{5}_{0101}$ D_2	$^{6}_{0110}$ D_3	$^{7}_{0111}$ D_4
$^{8}_{1000}$ P_8	$^{9}_{1001}$ D_5	$^{10}_{1010}$ D_6	$^{11}_{1011}$ D_7
$^{12}_{1100}$ D_8	$^{13}_{1101}$ D_9	$^{14}_{1110}$ D_{10}	$^{15}_{1111}$ D_{11}

P2=D1⊕ D3 ⊕ D4 ⊕ D6⊕ D7⊕ D10⊕ D11

	$^{1}_{0001}$ P_1	$^{2}_{0010}$ P_2	$^{3}_{0011}$ D_1
$^{4}_{0100}$ P_4	$^{5}_{0101}$ D_2	$^{6}_{0110}$ D_3	$^{7}_{0111}$ D_4
$^{8}_{1000}$ P_8	$^{9}_{1001}$ D_5	$^{10}_{1010}$ D_6	$^{11}_{1011}$ D_7
$^{12}_{1100}$ D_8	$^{13}_{1101}$ D_9	$^{14}_{1110}$ D_{10}	$^{15}_{1111}$ D_{11}

P4=D2⊕ D3 ⊕ D4 ⊕ D8⊕ D9⊕ D10⊕ D11

	$^{1}_{0001}$ P_1	$^{2}_{0010}$ P_2	$^{3}_{0011}$ D_1
$^{4}_{0100}$ P_4	$^{5}_{0101}$ D_2	$^{6}_{0110}$ D_3	$^{7}_{0111}$ D_4
$^{8}_{1000}$ P_8	$^{9}_{1001}$ D_5	$^{10}_{1010}$ D_6	$^{11}_{1011}$ D_7
$^{12}_{1100}$ D_8	$^{13}_{1101}$ D_9	$^{14}_{1110}$ D_{10}	$^{15}_{1111}$ D_{11}

P8=D5⊕ D6 ⊕ D7 ⊕ D8⊕ D9⊕ D10⊕ D11

Let for an example, our 11 bits message signal be,

D_{11}	D_{10}	D_9	D_8	D_7	D_6	D_5	D_4	D_3	D_2	D_1
1	0	0	1	1	1	0	1	0	0	1

	$^{1}_{0001}$ P_1	$^{2}_{0010}$ P_2	1
$^{4}_{0100}$ P_4	0	0	1
$^{8}_{1000}$ P_8	0	1	1
1	0	0	1

	0	0010 2 P_2	1
0100 4 P_4	0	0	1
1000 8 P_8	0	1	1
1	0	0	1

P_1= D_1⊕ D_2⊕ D_4 ⊕ D_5 ⊕ D_7⊕ D_9⊕ D_{11} = 0

	0	1	1
0100 4 P_4	0	0	1
1000 8 P_8	0	1	1
1	0	0	1

P2= D1⊕ D3 ⊕ D4 ⊕ D6⊕ D7⊕ D10⊕ D11
= 1

	0	1	1
1	0	0	1
1000 8 P_8	0	1	1
1	0	0	1

P4=D2⊕ D3 ⊕ D4 ⊕ D8⊕ D9⊕ D10⊕ D11
= 1

	0	1	1
1	0	0	1
0	0	1	1
1	0	0	1

P8=D5⊕ D6 ⊕ D7 ⊕ D8⊕ D9⊕ D10⊕ D11

= 0

So, total transferred bits are:

D_{11}	D_{10}	D_9	D_8	D_7	D_6	D_5	P_8	D_4	D_3	D_2	P_4	D_1	P_2	P1
1	0	0	1	1	1	0	0	1	0	0	1	1	1	0

	0	1	1
1	0	0	1
0	0	1	1
1	0	0	1

Suppose, for noisy channel, error has occurred at position 5.

D_{11}	D_{10}	D_9	D_8	D_7	D_6	D_5	P_8	D_4	D_3	D_2	P_4	D_1	P_2	P1
1	0	0	1	1	1	0	0	1	0	1	1	1	1	0



	0	1	1
1	1	0	1
0	0	1	1
1	0	0	1

Now, the received data will be:

D_{11}	D_{10}	D_9	D_8	D_7	D_6	D_5	P_8	D_4	D_3	D_2	P_4	D_1	P_2	P1
1	0	0	1	1	1	0	0	1	0	1	1	1	1	0

To find whether there is any error or not, we have to define 4 new variables. Let them be xP1, xP2, xP4, xP8

xP1 will check the region of P1 along with P1.

xP2, xP4, xP8 will also work in the same manner.

Therefore,

xP1= P1⊕ D1⊕ D2⊕ D4 ⊕ D5 ⊕ D7⊕ D9⊕ D11=1

	0	1	1
1	1	0	1
0	0	1	1
1	0	0	1

If the value of xP1 is '1', then there will be odd number of '1' present in the highlighted region. But, the number of '1' should be even.

Here, the value of xP1 represents that an error has occurred in that region and the binary representation of the position where error has occurred has the form of "X X X 1"

Again,

xP2= P2⊕ D1⊕ D3 ⊕ D4 ⊕ D6⊕ D7⊕ D10⊕ D11=0

Here, the value of xP2 represents that there is no error in that region. So, the place where the error has occurred has not the binary representation of the form "X X 1 X".

Therefore, binary representation of the place where the error has occurred has '0' in the second position.

	0	1	1
1	1	0	1
0	0	1	1
1	0	0	1

Similarly,

xP4= P4⊕ D2⊕ D3 ⊕ D4 ⊕ D8⊕ D9⊕ D10⊕ D11=1

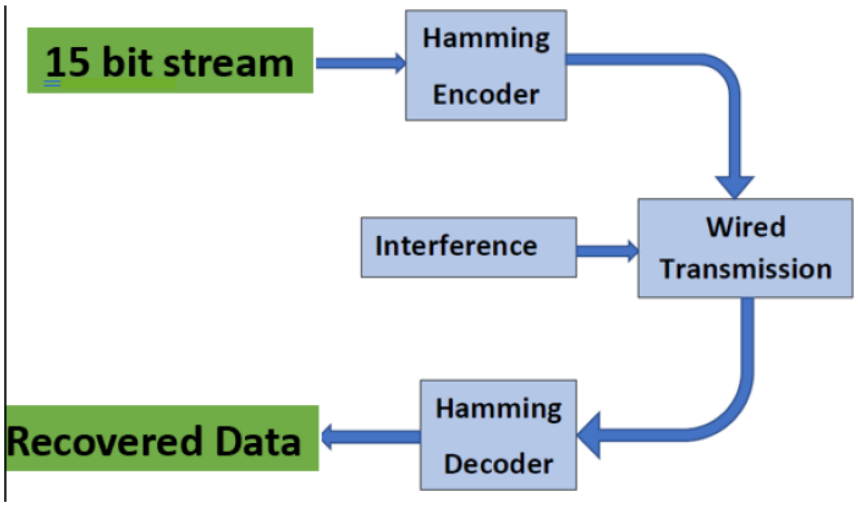
xP8= P8⊕ D5⊕ D6 ⊕ D7 ⊕ D8⊕ D9⊕ D10⊕ D11=0

xPi	Value	Comment on the binary representation of the place where error has occurred
xP_1	1	has '1' at the 1st bit
xP_2	0	has '0' at the 2nd bit
xP_4	1	has '1' at the 3rd bit
xP_8	0	has '0' at the 4th bit

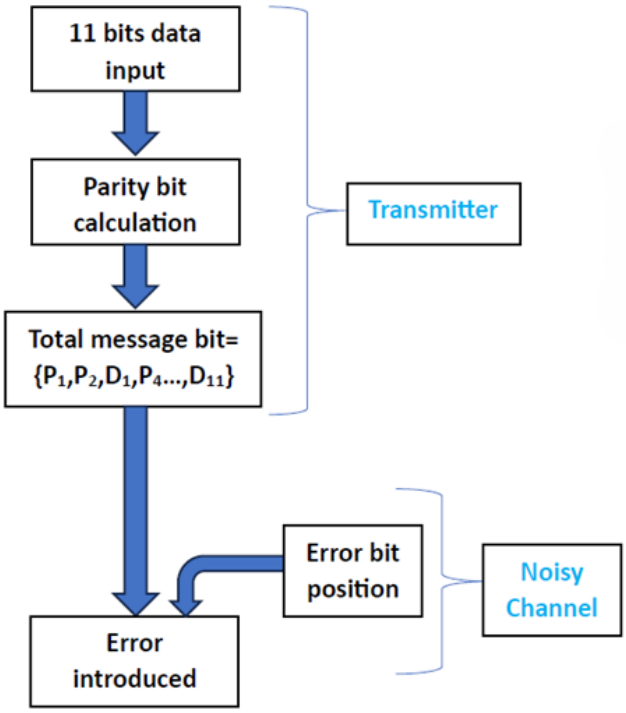
Now, if we combine all the four comments or we place the value of all xP1, xP2, xP4, xP8 serially, we will get the binary representation of the place where error has occurred. Here, that is **(1001)2=(5)10**

3.2 Design Method (PO(a))

Flow Diagram of a (15,11,4) Hamming code system:



Block Diagram:



The learner will be able to select a group of inputted data separately to calculate the parity bits (P_1, P_2, P_4, P_8) and after calculating the parity bit he/she also can justify his/her answers instantly.

The learner will be able to select a group of received data to calculate the errored parity bits (xP_1, xP_2, xP_4, xP_8) and after calculating the parity bit he/she also can justify his/her answers instantly.

After calculating the xP1, xP2, xP4, xP8, one can easily identify the position where error occurred and then that error can be corrected by just toggling that bit. So, in this CPLD module we just try to help the learner upto calculating xP1, xP2, xP4, xP8.

4.Implementation:

Inputs and Outputs:

Pin Planner:

Node Name	Location	Node Name	Location
LED_COM	PIN_141	data_inp[7]	PIN_72
X_parity_bit[1]	PIN_20	data_inp[8]	PIN_73
X_parity_bit[2]	PIN_19	data_inp[9]	PIN_78
X_parity_bit[3]	PIN_18	data_inp[10]	PIN_79
X_parity_bit[4]	PIN_17	data_inp[11]	PIN_80
clk	PIN_55	error_place[1]	PIN_82
col_green[1]	PIN_112	error_place[2]	PIN_83
col_green[2]	PIN_113	error_place[3]	PIN_86
col_green[3]	PIN_114	error_place[4]	PIN_87
col_green[4]	PIN_116	parity_corr_bit_disp[1]	PIN_51
col_red[1]	PIN_98	parity_corr_bit_disp[2]	PIN_49
col_red[2]	PIN_99	parity_corr_bit_disp[3]	PIN_48
col_red[3]	PIN_100	parity_corr_bit_disp[4]	PIN_47
col_red[4]	PIN_101	reset	PIN_63
data_inp[1]	PIN_64	row[1]	PIN_88
data_inp[2]	PIN_65	row[2]	PIN_89
data_inp[3]	PIN_67	row[3]	PIN_90
data_inp[4]	PIN_68	row[4]	PIN_91
data_inp[5]	PIN_69	with_error	PIN_60
data_inp[6]	PIN_70	with_parity	PIN_62

4.1 Description:

X_parity_bit[4:1]	They are used to display the errored parity bits through 4 LEDs.
clk	System clock
col_green[1:4]	To control the green led column of 4x4 LED matrix
col_red[1:4]	To control the red led column of 4x4 LED matrix
data_inp[1:11]	To provide inputs or data bits to transmit
error_place[1:4]	To input the place where error will be occurred
parity_corr_bit_disp[1:4]	Buttons to select which group of data learner want to see
reset	Reset button
row[1:4]	To control the rows of 4x4 LED matrix
with_error	To introduce error into the message bits
with_parity **(Answer Button)**	To justify the answer of learner's calculation

Steps to use this Learning Module:

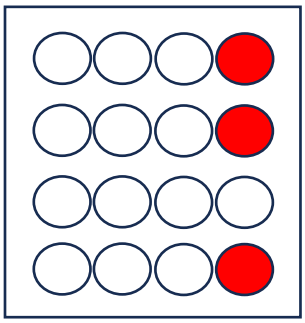
- Firstly, earner should provide the 11 bits input and the place where the error will be introduced through 15 DIP switches. Reset button (PIN 63) should be 'OFF' for normal operation.

  Let the data bits we want to transmit be

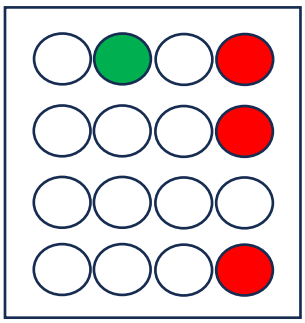
D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11
1	0	1	1	0	0	0	1	0	1	1

  And the error place is 6 (0110)

- Now, if the learner press sw4, then bits corresponding to P_1 will be shown. If we look at LED matrix, then we will see something like this:

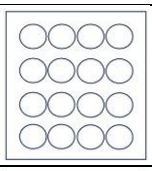
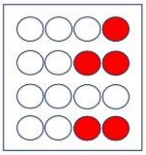
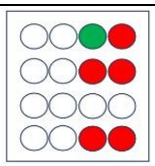


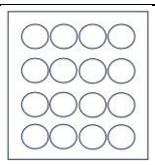
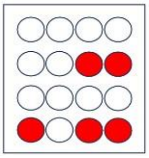
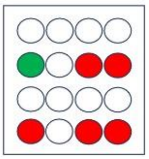
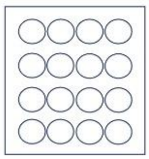
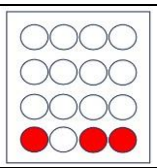
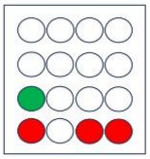
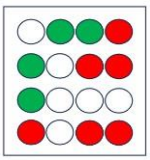
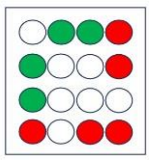
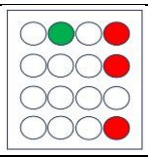
Now, the learner can evaluate the value of P_1. As here, the number of 1 is odd, P_1= 1. After this, the learner should press sw7 to validate his/her answer. The LED matrix should be like this:

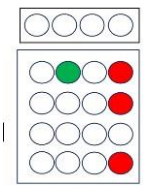


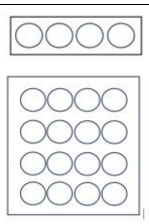
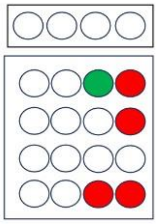
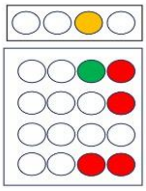
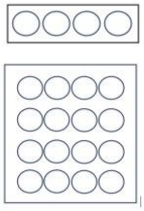
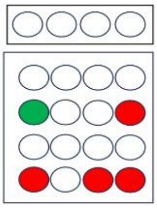
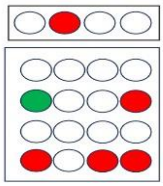
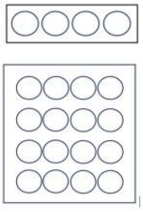
The green light indicates P_1=1.

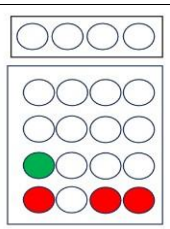
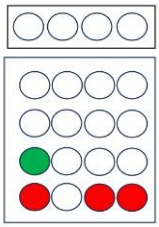
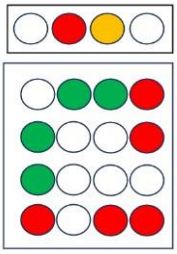
After this point, how the learner should proceed is given below in a table: (corresponding LED matrix has also been provided)

• Sw7=0, sw4=0 (turned off)	
• Sw3=1	
• Sw7=1	

• Sw7=0, sw3=0	
• Sw2=1	
• Sw7=1	
• Sw7=0, sw2=0	
• Sw1=1	
• Sw7=1	
• Sw1=sw2=sw3=sw4=sw7=1	
• Sw6=1, sw7=0	
• sw4=1, Sw1=sw2=sw3=0	

• sw7=1	

• Sw7=0, sw4=0	
• sw3=1	
• sw7=1	
• Sw7=0, sw3=0	
• Sw2=1	
• sw7=1	
• Sw7=0, sw2=0	

• Sw1=1	
• sw7=1	
• Sw1=sw2=sw3=sw4=sw7=1	

Here, the 4 LEDs represent that error has introduced at bit position $(110)_2=(6)_{10}$
Similarly, our CPLD module can be used to realize the (15,11,4) Hamming code system for any message bits.

Part: 02

Visualizing (31,26,5) Hamming Code system in FPGA:

The second part of the project focuses on visualizing the 31-bit Hamming code, comprising 26 data bits and 5 parity bits, using an FPGA. This implementation demonstrates the encoding, error detection, and correction processes in real time. Data input is managed through FPGA switches, while seven-segment displays showcase the original data, parity bits, generated Hamming code, and corrected output. This visualization provides an interactive way to understand the internal workings of Hamming code-based error correction, emphasizing its significance in ensuring data reliability in communication systems.
Here, total number of data bits will be 31,
Number of message bits will be

26, And there will be 5 parity

bits.

Extra parity bit will be at 16th bit position.

Now, if $\mathbf{xP_{16}}$ at the receiver side is 1, then we will be confirmed that the error is located anywhere between 16-to-31-bit position. And, if $\mathbf{xP_{16}}$ is 0, then the error will be at between 1-

to-15-bit position.
Then the correct error location can be traced by using other 4 parity bits.
In fact, we can say, two (15,11,4) Hamming code system creates one (31,26,5) Hamming code system.

Details of our FPGA based (31,26,5) Hamming Code System:

FPGA Board Model:

In this project, we have used the ***Terasic DE1-SoC board***, which features the powerful Cyclone V SoC 5CSEMA5F31 device.
This board integrates 85,000 programmable logic elements and 4,450 Kbits of embedded memory, offering a robust platform for implementing complex digital systems. The combination of FPGA fabric with an ARM-based hard processor system (HPS) makes it ideal for applications that require high performance, flexibility, and real-time processing. Its versatility and advanced features make the DE1-SoC board an excellent choice for FPGAbased error correction systems, such as the 31-bit Hamming code in this project.

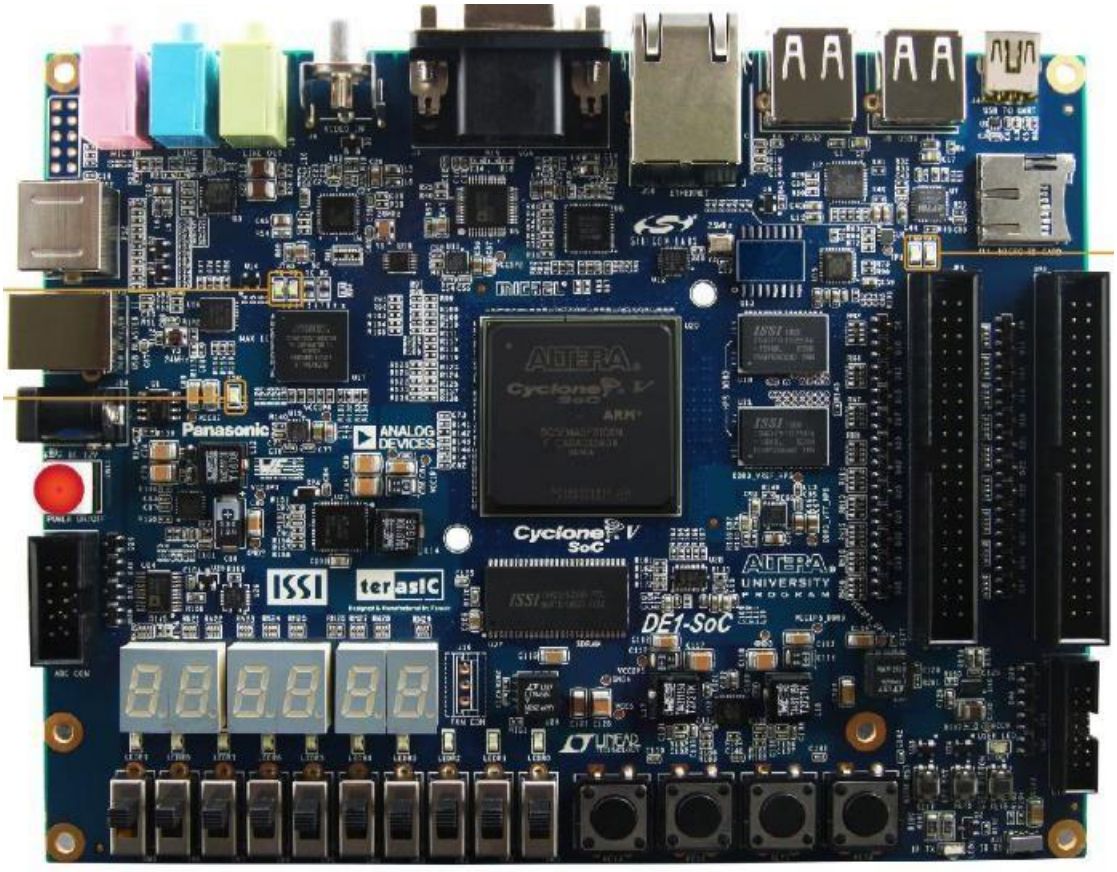


Fig: Terasic DE1-SoC board, with Cyclone V SoC 5CSEMA5F31 device.

Input Taking:

To provide the inputs we will use 10 slide switches and 4 push-buttons built in on the FPGA board. And another important thing to be mentioned is that we will provide input in **decimal value** as providing 26 input bits in binary value (bit by bit) is not an easy task.

10 Slide Switches:

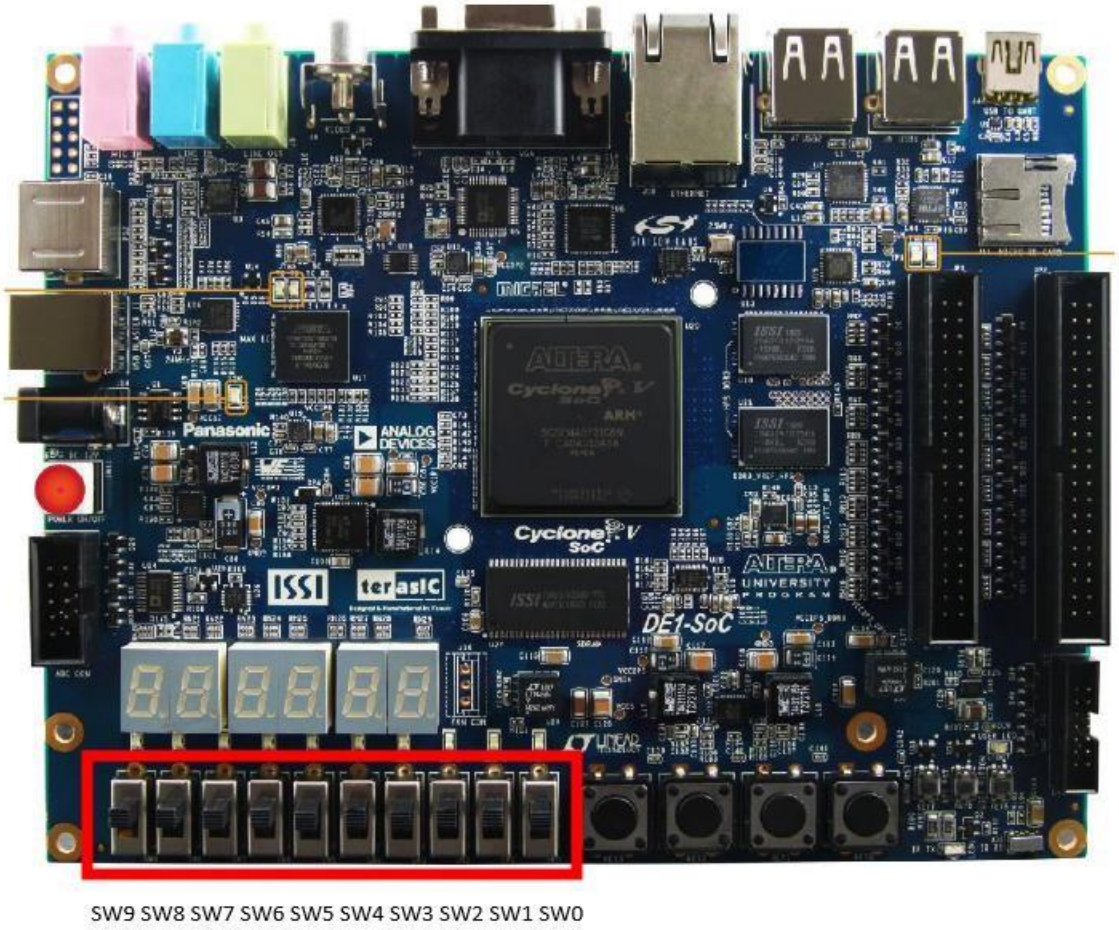


Fig: 10 slide switches

Debouncing Slide switches:

Though using slide switches without debouncing is not erroneous as using pushbuttons without debouncing, we debounce the slide switches. For that we will use a simple D flip-flop module in our Verilog code and a slowed clock pulse (4 kHz) to make the activity of the slide switches synchronous.

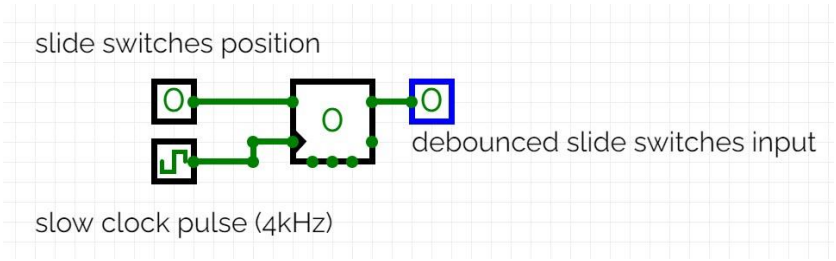


Fig: circuit for debouncing slide switches

(SW9) ten power button:

1 (ON)	Inputs will be changed by 10^7, 10^6, 10^5 or 10^4
0 (OFF)	Inputs will be changed by 10^3, 10^2, 10^1 or 10^0

(SW8) plus minus button:

1 (ON)	Inputs will be increased (plus)
0 (OFF)	Inputs will be decreased (minus)

(SW7) data error button:

1 (ON)	The place (bit position) where error will be occurred taken as input (in decimal value)
0 (OFF)	Input 26 bits will be taken as input (in decimal value)

(SW6) input display button:

1 (ON)	Inputs will be shown in the 7-segment display in its decimal value
0 (OFF)	Inputs will not be shown

(SW5) error place display button:

1 (ON)	The place where error will occur will be shown in the 7-segment display
0 (OFF)	Error position will not be shown

(SW4) input with parity display button:

1 (ON)	Message 31 bits (26 input bits + 5 parity bits) will be displayed on the 7-segment display in its decimal value
0 (OFF)	Message bits will not be shown

(SW3) after error display button:

1 (ON)	Message 31 bits after occurring the error in the desired position will be displayed on the 7-segment display in its decimal value
0 (OFF)	Errored message bits will not be shown

(SW2) after correction display button:

1 (ON)	Message 31 bits after correcting the error will be displayed on the 7-segment display in its decimal value
0 (OFF)	corrected message bits will not be shown

NB: SW6 to SW2 has a priority list of SW6> SW5> SW4> SW3> SW2.

That means, if SW6 is in its 'ON' position then the position of other switches will be ignored and input 26 bits will be displayed in its decimal value on the 7-segment display. Similarly, if SW5 is in its 'ON' position and SW6 is in its 'OFF' position, then activity for the active position of SW5 will occur and position of SW4 to SW2 will be ignored. SW4 to SW2 will also work in the same manner.

(SW1) RESET button:

1 (ON)	Input 26 bits, error place will be reset. (all the bits will be 0 and decimal value will be 0 also)
0 (OFF)	Normal operation of the system will continue

(SW0) MSD or LSD button:

1 (ON)	It will display MSD to 7th digit for the value which is displayed in the 7-segment display
0 (OFF)	It will display 6th to 1st (LSD) digit for the value which is displayed in the 7-segment display

(MSD- Most Significant Digit, LSD- Least Significant Digit) **Push Buttons:**

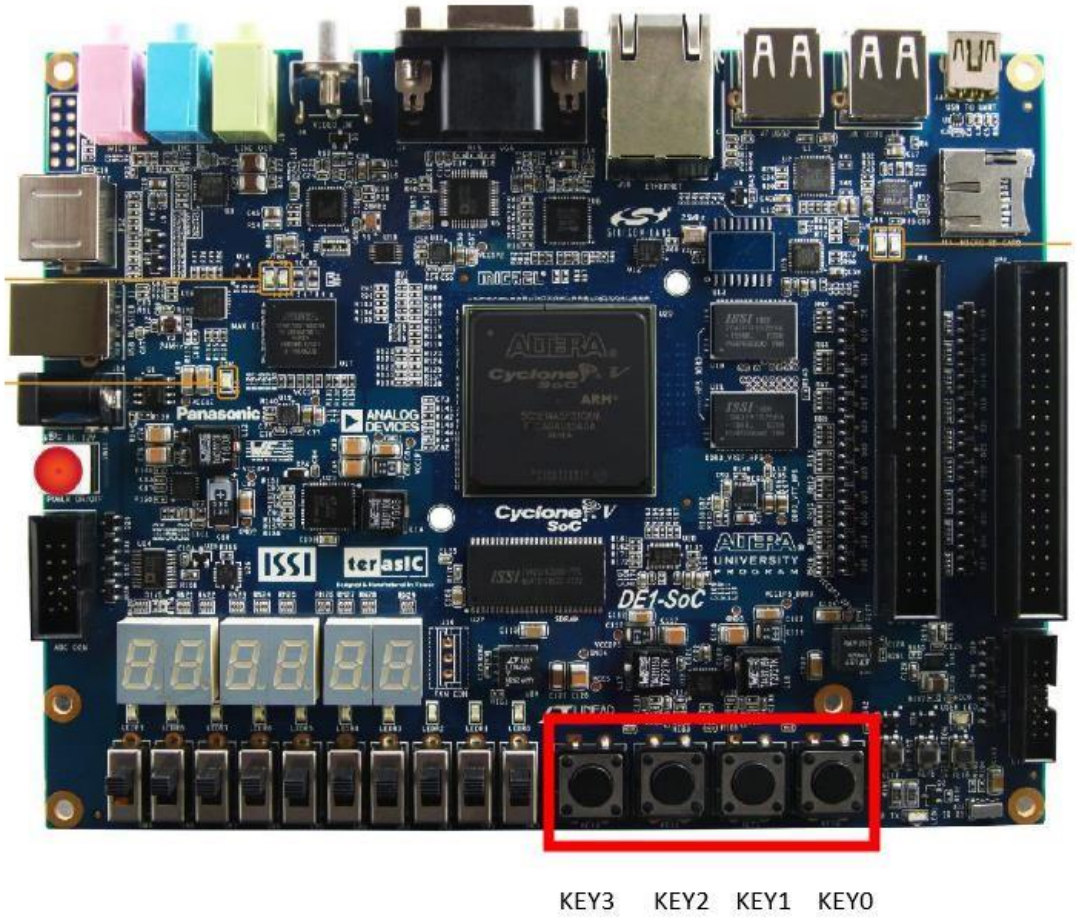


Fig: Four push buttons

Debouncing push buttons:

Sometimes, using push buttons without accounting the effect of 'debouncing' may cause erroneous result. So, in our project we will use a module in our Verilog code to debounce the button pushing.

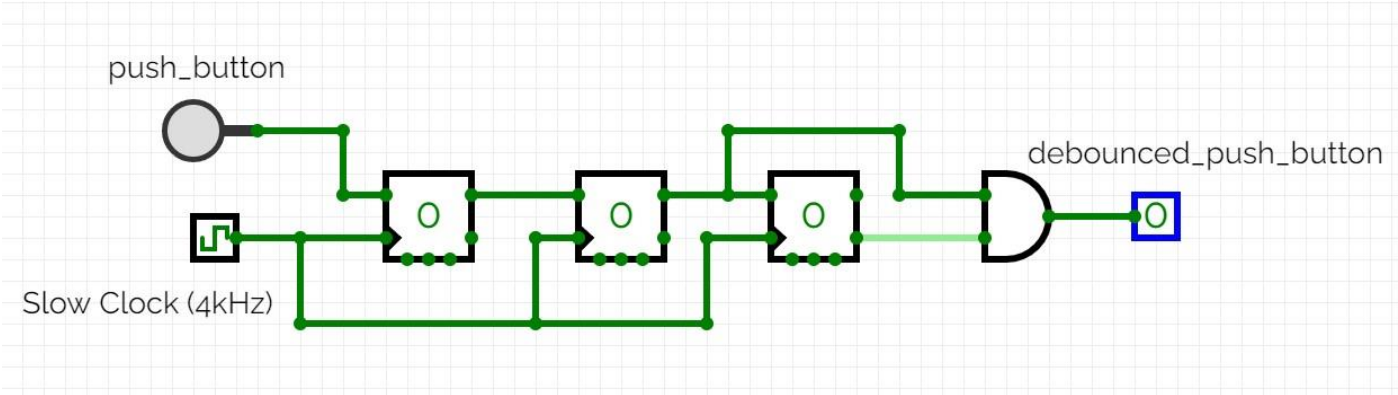


Fig: circuit for debouncing push-buttons

Activity of push buttons:

Push Buttons	Data Error button	plus minus button	ten power button	Result
KEY3	OFF	ON	ON	Input will be increased by 10^7
	OFF	ON	OFF	Input will be increased by 10^3
	OFF	OFF	ON	Input will be decreased by 10^7
	OFF	OFF	OFF	Input will be decreased by 10^3
	ON	ON	X	*Nothing!!!*
	ON	OFF	X	*Nothing!!!*
KEY2	OFF	ON	ON	Input will be increased by 10^6
	OFF	ON	OFF	Input will be increased by 10^2
	OFF	OFF	ON	Input will be decreased by 10^6
	OFF	OFF	OFF	Input will be decreased by 10^2
	ON	ON	X	*Nothing!!!*
	ON	OFF	X	*Nothing!!!*

Push Buttons	Data Error button	plus minus button	ten power button	Result
KEY1	OFF	ON	ON	Input will be increased by 10^5
	OFF	ON	OFF	Input will be increased by 10^1
	OFF	OFF	ON	Input will be decreased by 10^5
	OFF	OFF	OFF	Input will be decreased by 10^1
	ON	ON	X	*Error will be increased by 10^1*
	ON	OFF	X	*Error will be decreased by 10^1*
KEY0	OFF	ON	ON	Input will be increased by 10^6
	OFF	ON	OFF	Input will be increased by 10^2
	OFF	OFF	ON	Input will be decreased by 10^6
	OFF	OFF	OFF	Input will be decreased by 10^2
	ON	ON	X	*Error will be increased by 10^0*
	ON	OFF	X	*Error will be decreased by 10^0*

Clock Input:

For this project we will use the built-in 50MHz clock pulses from our FPGA board. Pin number for the clock pulse we will use is ‘AA16’.

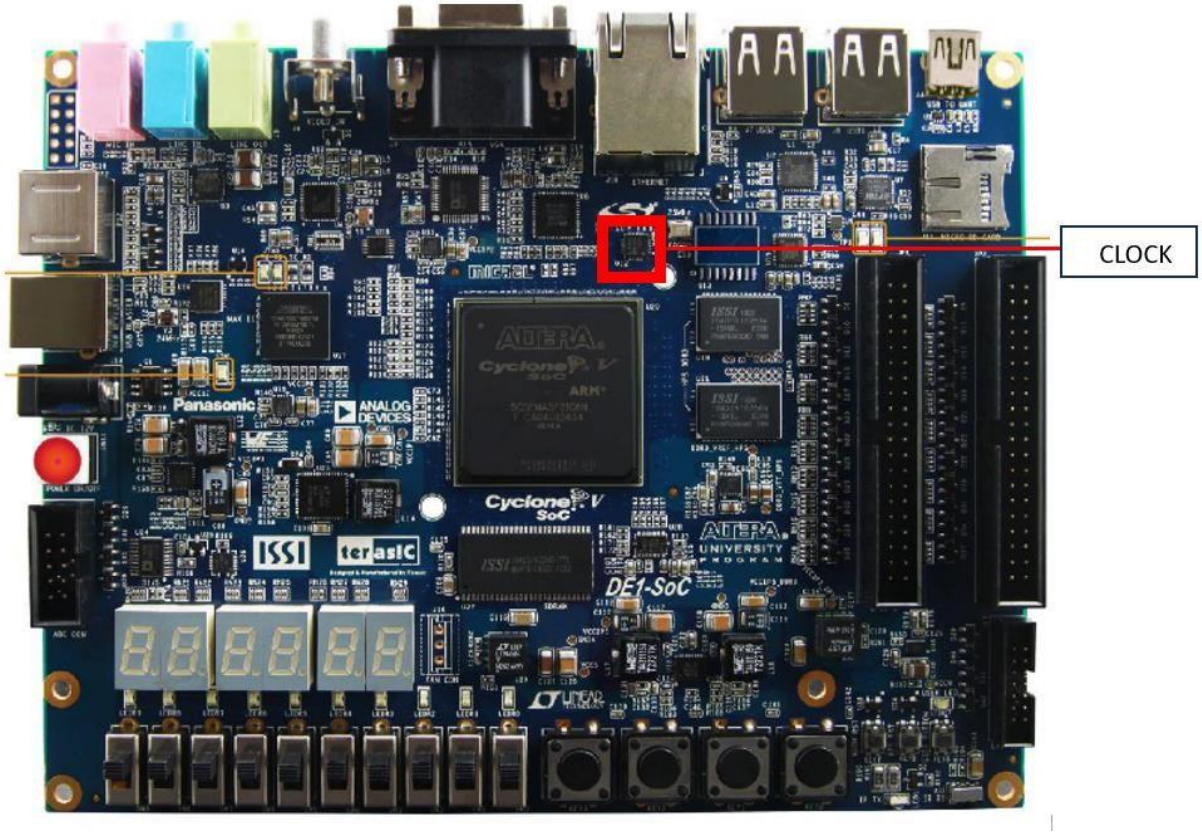


Fig: Clock pulse component in our FPGA board

Output in Six 7-segment Displays:

For our project, we will display our output in decimal value and to display the digits we will use built in six 7-segment displays. As there are 6 digit displaying capability, we will display 6 digits at once. If there are more than 6 digits in the output, we have to use the slide switch 'MSD or LSD button' to select the part of the output which we want to observe (either MSD to 7th digit or 6th to LSD)

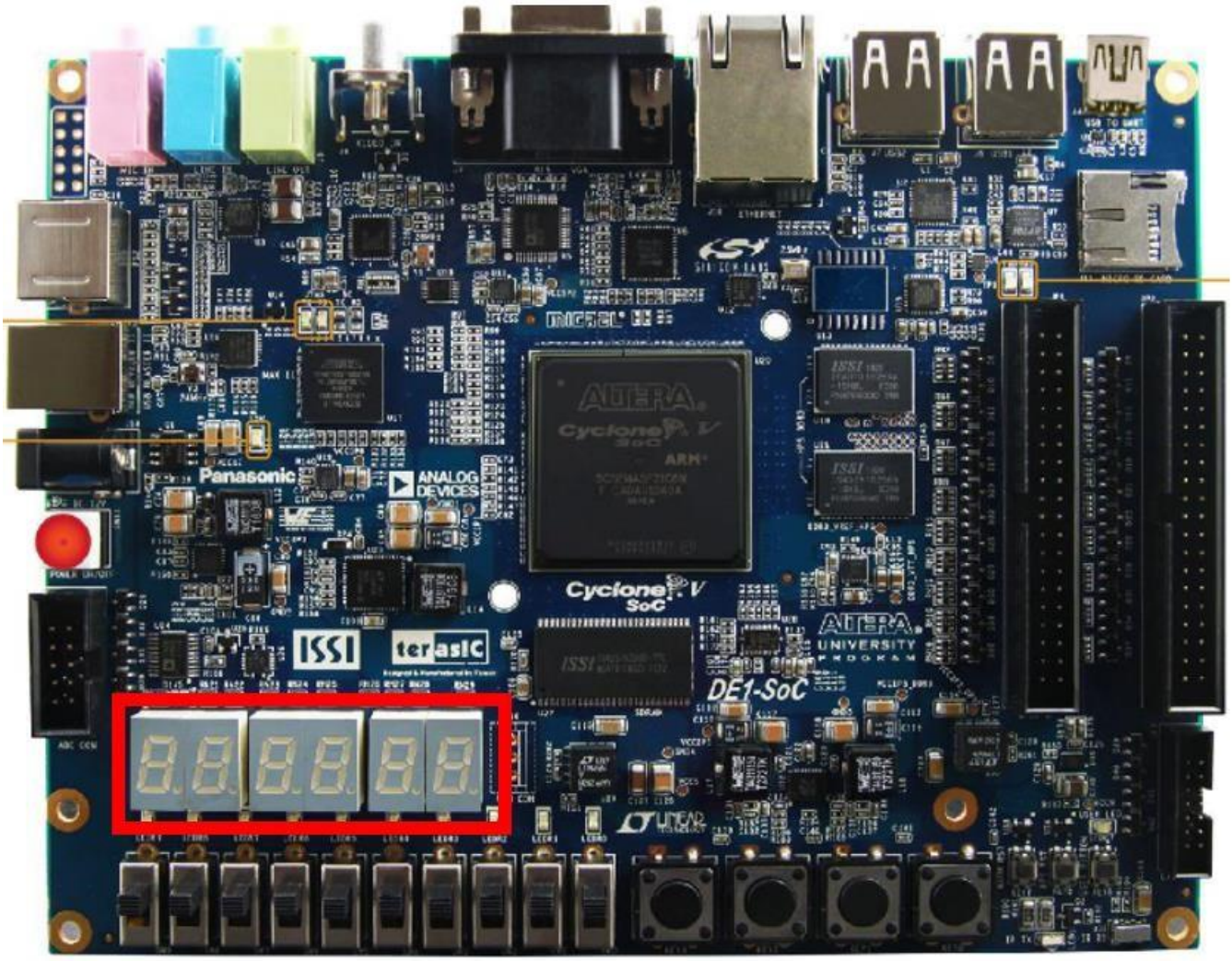
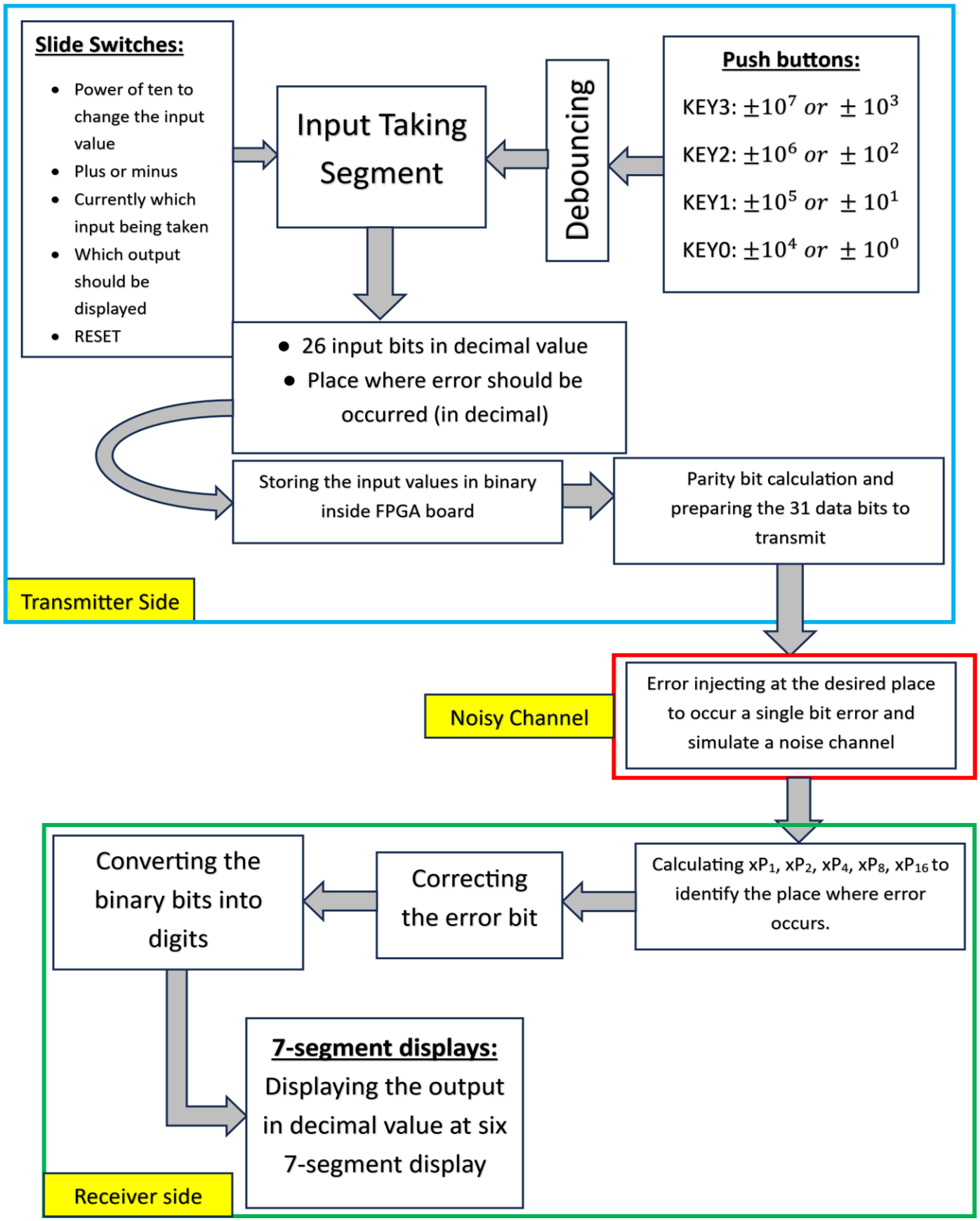


Fig: Six built in 7-segment displays of our FPGA board

Flow diagram of our FPGA based (31,26,5) Hamming code system:


Slide Switches:
Power of ten to change the input value
Plus or minus
Currently which input being taken
Which output should be displayed
RESET
Input Taking Segment
Debouncing
Push buttons:
KEY3: $\pm 10^7$ or $\pm 10^3$
KEY2: $\pm 10^6$ or $\pm 10^2$
KEY1: $\pm 10^5$ or $\pm 10^1$
KEY0: $\pm 10^4$ or $\pm 10^0$
26 input bits in decimal value
Place where error should be occurred (in decimal)
Storing the input values in binary inside FPGA board
Parity bit calculation and preparing the 31 data bits to transmit
Transmitter Side
Noisy Channel
Error injecting at the desired place to occur a single bit error and simulate a noise channel
Calculating xP_1, xP_2, xP_4, xP_8, xP_{16} to identify the place where error occurs.
Correcting the error bit
Converting the binary bits into digits
7-segment displays:
Displaying the output in decimal value at six 7-segment display
Receiver side

5. Design Analysis and Evaluation

5.1 Novelty

The project introduces a real-time visualization of the Hamming code error correction process using an FPGA and CPLD platform. By combining the educational aspects with real-world applications, this dual-platform approach distinguishes itself from conventional error correction demonstrations. The inclusion of a hands-on learning module for the (15,11,4) Hamming code system enhances its appeal as an educational tool.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to Public Health and Safety

The system prioritizes safe usage by employing error-detection algorithms that ensure data integrity, particularly in critical applications like medical data transmission or industrial control systems. For example, accurate transmission of vital signs or command signals is achieved through robust error correction mechanisms, minimizing risks in life-critical scenarios.

5.2.2 Considerations to Environment

The FPGA and CPLD modules were chosen for their energy efficiency, aligning with modern sustainable design principles. By visualizing error correction processes efficiently, the system reduces energy consumption associated with debugging or error-prone transmissions in larger systems.

5.2.3 Considerations to Cultural and Societal Needs

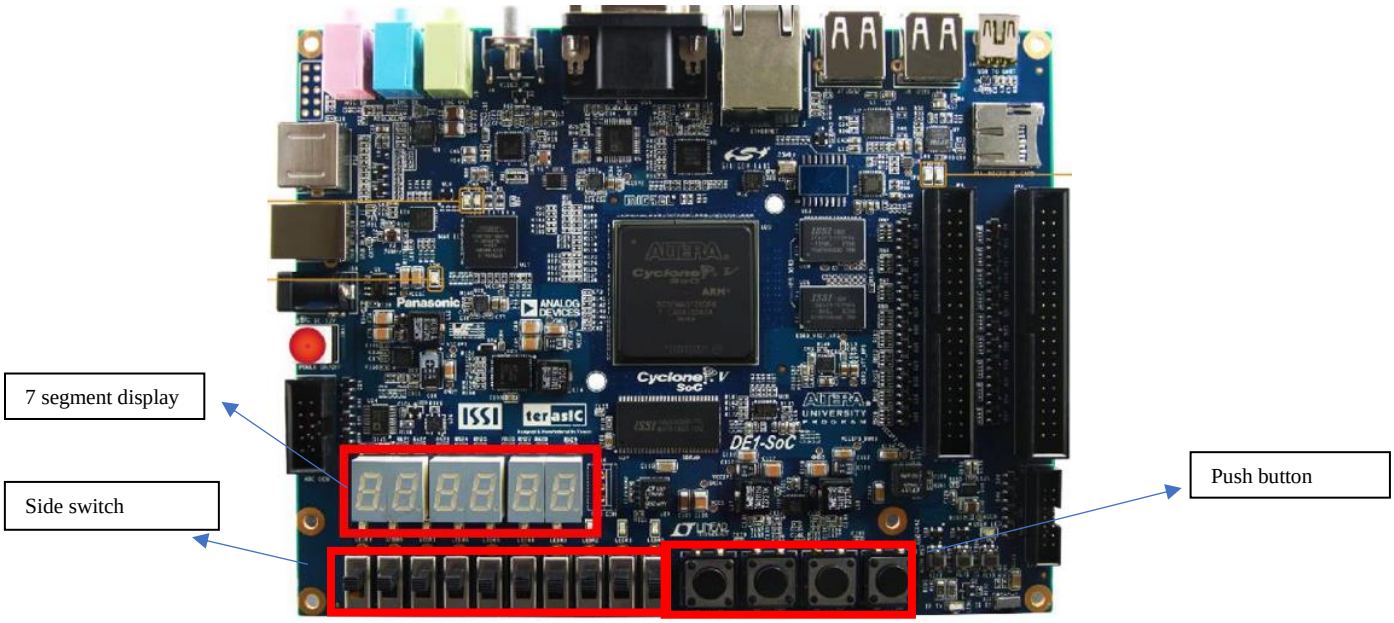
The project addresses societal needs by fostering technological literacy. It serves as an educational resource, particularly in regions where advanced communication systems are being adopted, contributing to workforce development and capacity building.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

The experiments were designed to verify the implementation of error detection and correction for predefined test cases. We made a learning module in the CPLD Board for the 15-bit hamming code. Also made a real-time visualization of the Hamming code error correction process using an FPGA. In the FPGA Board we use the 7-segment display, slide switch , puch button. Input was systematically varied to simulate realistic data transmission scenarios, including intentional error

introduction to evaluate system accuracy.



5.3.2 Data Collection

Test cases to show input and output of our project:

Test Case: 01

Suppose our 26 bits input is:
In decimal:
63201247 In
binary (26 bits):

1 1	1 1 0 0	0 1 0 0	0 1 0 1	1 1 1 1	1 1 0 1	1 1 1 1

And, we want to occur an error at the bit position 21.

In binary (5 bits): 10101

Then, by calculating the parity bits and place them in the correct place in a MATLAB code, we found that total 31 message bits will be:

In binary (31 bits):

111	1000	1000	1011	0111	1101	1111	0110

In decimal: **2022407670**

Then, if we introduce an error at the 21th bit position, then the total 31 message bits will be:

In binary (31 bits):

111	1000	1001	1011	0111	1101	1111	0110

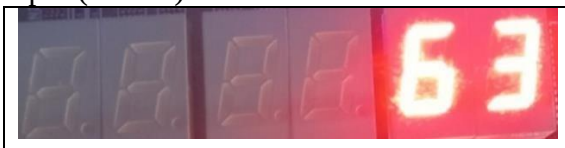
In decimal: **2023456246**

So, the corrected 31 data bits should be: (similar to the input with parity bits) In binary (31 bits):

111	1000	1000	1011	0111	1101	1111	0110

In decimal: **2022407670**

Now, let's see how our FPGA based system handles this case:
Input (26 bits) in decimal value: 63201247





Desired error place in decimal value: 21



Total 31 message bits in decimal value (transmitted data bits) **2022407670**





31 message bits in decimal value after error occurs: **2023456246**





31 message bits in decimal value after correction: (same as the transmitted data bits) **2022407670**





Our project will display output on the basis of provided input on the six 7-segment display like this way.

MATLAB Code result for Test Case: 01

```
Input 26 bits data in decimal value:
    63201247

Input 26 bits data in binary:
1  1  1  1  0  0  0  1  0  0  0  1  0  1  1  1  1  1  1  1  0  1  1  1  1  1

31 bits data in binary after parity bit calculation:
1  1  1  1  0  0  0  1  0  0  0  1  0  1  1  0  1  1  1  1  1  0  1  1  1  1  1  0  1  1  0

31 bits data in decimal value after parity bit calculation:
               2022407670

errored data in binary value:
1  1  1  1  0  0  0  1  0  0  1  1  0  1  1  0  1  1  1  1  1  0  1  1  1  1  1  0  1  1  0

errored data in decimal value:
               2023456246
```

Which perfectly matches with the obtained result.

Test case :02

Input (26 bits) in decimal value: 07523184




Desired error place in decimal value:12



Total 31 message bits in decimal value (transmitted data bits) : 0240760577




31 message bits in decimal value after error occurs: 0240762625




31 message bits in decimal value after correction (same as the transmitted data bits): 0240760577




MATLAB Code Result for Test Case: 02

```
Input 26 bits data in decimal value:
     7523184

Input 26 bits data in binary:
0  0  0  1  1  1  0  0  1  0  1  1  0  0  1  0  1  1  0  1  1  1  0  0  0  0

31 bits data in binary after parity bit calculation:
0  0  0  1  1  1  0  0  1  0  1  1  0  0  1  1  0  1  1  0  1  1  1  0  0  0  0  0  0  0  1

31 bits data in decimal value after parity bit calculation:
   240760577

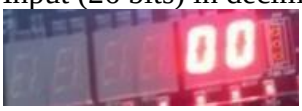
errored data in binary value:
0  0  0  1  1  1  0  0  1  0  1  1  0  0  1  1  0  1  1  1  1  1  1  0  0  0  0  0  0  0  1

errored data in decimal value:
   240762625
```

Which also perfectly matches with the obtained result.

Test Case: 03

Input (26 bits) in decimal value: 00096494




Desired error place in decimal value: 29



Total 31 message bits in decimal value (transmitted data bits): 003116787




31 message bits in decimal value after error occurs: 0271552243




31 message bits in decimal value after correction (same as the transmitted data bits): 0003116787




MATLAB Code Result for Test case: 03

```
Input 26 bits data in decimal value:
       96494

Input 26 bits data in binary:
0  0  0  0  0  0  0  0  0  1  0  1  1  1  1  0  0  0  1  1  1  0  1  1  1  0

31 bits data in binary after parity bit calculation:
0  0  0  0  0  0  0  0  0  1  0  1  1  1  1  1  0  0  0  1  1  1  0  1  1  1  1  0  0  1  1

31 bits data in decimal value after parity bit calculation:
     3116787

errored data in binary value:
0  0  1  0  0  0  0  0  0  1  0  1  1  1  1  1  0  0  0  1  1  1  0  1  1  1  1  0  0  1  1

errored data in decimal value:
   271552243
```

Which also perfectly matches with the obtained result

Test Case: 04

Input (26 bits) in decimal value: 00 000 129




Desired error place in decimal value:03



Total 31 message bits in decimal value (transmitted data bits):000 000 2191



31 message bits in decimal value after error occurs: 000 000 2187




31 message bits in decimal value after correction (same as the transmitted data bits): 000 000 2191




MATLAB Code result for test case: 04

```
Input 26 bits data in decimal value:
   129

Input 26 bits data in binary:
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  1

31 bits data in binary after parity bit calculation:
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  1  0  0  0  1  1  1  1

31 bits data in decimal value after parity bit calculation:
        2191

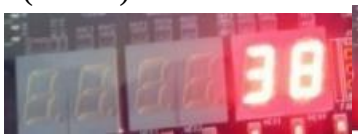
errored data in binary value:
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  1  0  0  0  1  0  1  1

errored data in decimal value:
        2187
```

Which also perfectly matches with the obtained result

Test Case: 05

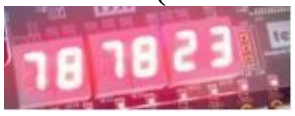
Input (26 bits) in decimal vale: 38212653



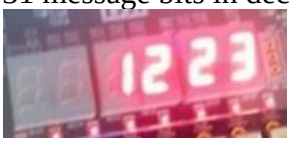

Desired error place in decimal value: 19



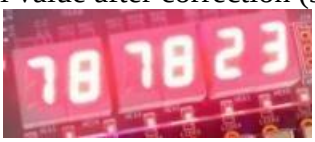
Total 31 message bits in decimal value (transmitted data bits): 1222787823

31 message bits in decimal value after error occurs: 1223049967




31 message bits in decimal value after correction (same as the transmitted data bits): 1222787823

MATLAB Code result for test case: 05

```
Input 26 bits data in decimal value:
    38212653

Input 26 bits data in binary:
1  0  0  1  0  0  0  1  1  1  0  0  0  1  0  1  0  0  0  0  1  0  1  1  0  1

31 bits data in binary after parity bit calculation:
1  0  0  1  0  0  0  1  1  1  0  0  0  1  0  0  1  0  0  0  0  1  0  1  1  1  0  1  1  1  1

31 bits data in decimal value after parity bit calculation:
                1222787823

errored data in binary value:
1  0  0  1  0  0  0  1  1  1  0  0  1  1  0  0  1  0  0  0  0  1  0  1  1  1  0  1  1  1  1

errored data in decimal value:
                1223049967
```

Which also perfectly matches with the obtained result

5.3.3 Results and Analysis

Results demonstrated that the system consistently corrected single-bit errors while accurately detecting error positions. For example, with an input decimal value of 63201247 and an error at position 21, the output was corrected to match the expected transmitted data.

MATLAB Code result for Test Case: 01

```
Input 26 bits data in decimal value:
    63201247

Input 26 bits data in binary:
1  1  1  1  0  0  0  1  0  0  0  1  0  1  1  1  1  1  1  1  0  1  1  1  1  1

31 bits data in binary after parity bit calculation:
1  1  1  1  0  0  0  1  0  0  0  1  0  1  1  0  1  1  1  1  1  0  1  1  1  1  1  0  1  1  0

31 bits data in decimal value after parity bit calculation:
               2022407670

errored data in binary value:
1  1  1  1  0  0  0  1  0  0  1  1  0  1  1  0  1  1  1  1  1  0  1  1  1  1  1  0  1  1  0

errored data in decimal value:
               2023456246
```

Which perfectly matches with the obtained result.

5.3.4 Interpretation and Conclusions on Data

The analysis confirmed the reliability of the FPGA and CPLD-based systems in real-time error correction. The visualized outputs were in complete agreement with MATLAB simulations, validating the design's functionality and theoretical accuracy.

5.4 Limitations of Tools (PO(e))

The FPGA platform, though robust, has a finite number of programmable logic elements, limiting the complexity of codes that can be implemented. Similarly, clock frequency constraints in the CPLD board restrict the speed of real-time applications. Minor tolerances in switch debouncing and potential errors due to hardware fatigue were noted.

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

The project promotes technical education, making complex error correction concepts accessible. It aligns with the broader societal goal of enhancing digital literacy and fostering innovation.

5.5.2 Assessment of Health and Safety Issues

By ensuring data integrity, particularly in health-monitoring devices and safety-critical systems, the project indirectly contributes to public safety.

5.5.3 Assessment of Legal Issues

All software and hardware used adhered to licensing requirements. Ethical guidelines for proper acknowledgment of external resources were followed.

5.6 Sustainability Evaluation (PO(g))

The use of low-power FPGA and CPLD modules aligns with sustainable practices, minimizing environmental impact. The project also advocates for reusable designs that can be adapted for future error-correction systems, reducing waste.

5.7 Ethical Issues (PO(h))

Ethical principles were upheld in all aspects of the project. Data integrity and transparency were prioritized, ensuring that results were accurate and reproducible. Challenges included ensuring proper citation of resources, which were mitigated by thorough referencing. Collaboration within the team was guided by mutual respect and equitable division of work.This draft can be further refined to fit specific course requirements or preferences. Let me know if adjustments are needed!

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

Student IDs	Contribution
2006099	Preparing Project Proposal, Testing the Verilog Code inside lab on FPGA Board for part 02, Verifying the test cases, Exploring the *Terasic DE1-SoC FPGA board,* Report Writing
2006117	Recording and verifying the test cases , Testing the Verilog Code inside lab on FPGA Board and CPLD board, Debugging the verilog code, Report Writing
2006118	Exploring the *Terasic DE1-SoC FPGA board,* Testing the Verilog Code inside lab on FPGA Board for part 02 , Debugging the verilog code, Report writing
2006121	Preparing Project Proposal, Preparing Updated Project Proposal, Exploring the *Terasic DE1-SoC FPGA board*, Writing the Verilog Code for part: 01 and 02, Testing the Verilog Code inside lab on FPGA Board for part 02, Testing the Verilog Code inside lab on CPLD Board for part 01, Recording and verifying the test cases, Report Writing

6.2 Mode of team work

SL no	Contribution	Student IDs
1	Preparing Project Proposal	2006099, 2006121
2	Preparing Updated Project Proposal	2006121
3	Exploring the *Terasic DE1-SoC FPGA board*	2006099, 2006118, 2006121
4	Writing the Verilog Code for part: 01 and 02	2006121
5	Testing the Verilog Code inside lab on FPGA Board for part 02	2006099,2006117, 2006118, 2006121
6	Testing the Verilog Code inside lab on CPLD Board for part 01	2006099,2006117, 2006118, 2006121
7	Recording and verifying the test cases	2006099,2006117, 2006121
8	Report Writing	2006099,2006117,2006118, 2006121

6.3 Diversity Statement of Team

Our team values diversity as a cornerstone of innovation and problem-solving. Comprising members from diverse academic backgrounds, technical expertise, and cultural perspectives, we brought together unique viewpoints to successfully implement the "31-bit Hamming Code" project using Verilog and FPGA. By embracing collaboration and inclusivity, we have fostered an environment where creative ideas thrive, ensuring a comprehensive approach to addressing challenges in digital communication and error correction. This diversity has empowered us to create robust, efficient solutions while appreciating the multifaceted experiences that each member contributes.

6.4 Log Book of Project Implementation

SL no.	Events	Date
1.	Preparing Project Proposal	29 September 2024
2.	Project Assigned	03 October 2024
3.	Started planning and lab work to complete the project	05 October 2024
4.	Completion of part-02	08 October 2024
5.	Recording test cases for part-02	08 October 2024
6.	Updated project proposal submission	08 October 2024
7.	First project update presentation	04 November 2024
8.	Part-01 assigned	11 November 2024
9.	Completed coding for part-01	13 November 2024
10.	Verifying and correcting the code for part-01 inside lab	20 November 2024
11.	Completion of part-01	20 November 2024
12.	Re-testing part-01	08 December 2024
13.	Report writing	09 December 2024

7. Communication to External Stakeholders (PO(j))

7.1 Executive Summary

This project successfully implements and visualizes Hamming code error correction through two integrated modules. The first module focuses on a 15-bit Hamming code learning system developed on a CPLD platform, offering an interactive educational experience. The second module involves a 31-bit Hamming code system implemented on an FPGA, capable of real-time error detection and correction for a 26-bit data input with 5 parity bits. User inputs are managed using switches and push buttons, while seven-segment displays show the encoded data, parity bits, and corrected outputs. This dual-platform approach highlights both theoretical and practical aspects of error correction, enhancing learning and demonstrating real-world applications.

7.2 User Manual

This project uses an FPGA board to implement a 31-bit and uses a CPLD board for 15 bit Hamming code for error detection and correction. It accepts a 26-bit input, adds 5 parity bits for FPGA Board and 11 bit input and 4 parity bits for CPLD board and corrects single-bit errors. Real-time visualization of encoding, error detection, and correction is provided.

Setup

1. **Hardware**:
    - Connect the FPGA board to your computer using a USB cable.
    - Ensure power is supplied to the FPGA board or CPLD board.
    - Connect a monitor or display for visualization (if required).
2. **Software**:
    - Program the FPGA with the provided bitstream using FPGA tools (e.g. Intel Quartus).
    - Use a serial terminal to interface with the system.

Operation

1. **Encoding**:
    - Enter a 26-bit data sequence via the serial terminal or FPGA input.
    - The system generates a 31-bit Hamming codeword with 5 parity bits.
2. **Error Injection** (Optional):
    - Introduce a single-bit error using the error injection feature on the terminal.

3. **Error Detection and Correction**:
    - The system identifies the erroneous bit position, corrects the error, and displays the corrected data.
4. **Visualization**:
    - The visualization interface shows the encoding, error detection, and correction process step by step.

Troubleshooting

Issue	Solution
No output on terminal	Check connections and FPGA programming.
Error not corrected	Ensure only single-bit errors are present.

7.3 Github Link

Verilog Code, pin-planner (Quartus project file) and MATLAB Code:

Google Drive Link:

https://drive.google.com/file/d/1OHwRH75VkNPEfwYhAsAcIdrtCj834NM5/view?usp=sharing

7.4 YouTube Link

Here video demonstration link of our project is given

Microsoft Onedrive Link:

- EEE_304_PROJECT

  " https://buetedu-my.sharepoint.com/:f:/g/personal/2006117_eee_buet_ac_bd/EhosY-viiwBNiIeawmNcXoEBd5b1Q2NtnO2PwyJUzKbxpQ?e=3VNW "

8 Project Management and Cost Analysis (PO(k))

8.1 Cost Analysis:

We use FPGA and CPLD board for this project which was provided by the lab. This project cost us nothing.

8.4 Timeline of Project Implementation:

SL no.	Events	Date
1.	Preparing Project Proposal	29 September 2024
2.	Project Assigned	03 October 2024
3.	Started planning and lab work to complete the project	05 October 2024
4.	Completion of part-02	08 October 2024
5.	Recording test cases for part-02	08 October 2024
6.	Updated project proposal submission	08 October 2024
7.	First project update presentation	04 November 2024
8.	Part-01 assigned	11 November 2024
9.	Completed coding for part-01	13 November 2024
10.	Verifying and correcting the code for part-01 inside lab	20 November 2024
11.	Completion of part-01	20 November 2024
12.	Re-testing part-01	08 December 2024
13.	Report writing	09 December 2024

9 Future Work

While the current project successfully implements and visualizes Hamming code error correction using FPGA and CPLD platforms, there are several directions for future enhancements and extensions:

1. Expansion to Multi-Bit Error Detection and Correction

The current implementation focuses on single-bit error correction. Future work can involve implementing more advanced error-correcting codes like BCH or Reed-Solomon codes that can detect and correct multiple-bit errors, increasing the system's robustness for noisy environments.

2. Integration with Real-Time Communication Systems

Integrating the FPGA-based system with real-world communication protocols like Ethernet, UART, or SPI can demonstrate its utility in live data transmission scenarios. This will provide insights into its performance in practical applications.

3. Optimizing Power Consumption

Investigating power optimization techniques for FPGA and CPLD implementations could further align the system with sustainable design goals. Dynamic clock management or power gating strategies could be explored to reduce energy usage.

4. Scalability for Larger Data Blocks

Future iterations can explore scaling the design for larger data blocks, such as 64-bit or 128-bit systems, to meet the needs of modern communication and storage systems that handle higher data rates.

5. Incorporating Machine Learning for Error Prediction

Integrating machine learning models to predict the likelihood and location of errors based on historical data patterns could enhance system efficiency and error correction precision.

6. Development of a User-Friendly Interface

Enhancing the current system with a GUI or touchscreen interface could make it more accessible for educational purposes and allow users to interact with the system more intuitively.

7. Portability and Miniaturization

Exploring implementation on more compact hardware platforms like System-on-Chip (SoC) or embedded microcontroller units (MCUs) could make the system portable for broader applications, including IoT devices.

8. Support for Adaptive Error Correction

Adding functionality for adaptive error correction, where the system dynamically adjusts the error-correcting algorithm based on the severity of noise or error rate in the channel, can make it more versatile in diverse operating conditions.

9. Collaborative and Cloud-Based Education Modules

Future work can include developing cloud-based simulation and collaboration tools for the Hamming code module, enabling remote learning and interactive demonstrations.

10. Application-Specific Customizations

Tailoring the system for specific industries, such as medical imaging, aerospace communication, or automotive systems, could demonstrate its practical benefits in specialized environments.

These directions not only extend the functionality of the current project but also align with evolving technological needs and educational objectives.

10 References:

1. Fundamentals of Digital Logic with Verilog Design by Stephen Brown, Zvonko Varensic (3rd edition)
2. Modern Digital and Analog Communication Systems by B.P. Lathi, Zhi Ding (4th Edition)
3. https://www.youtube.com/@3blue1brown
4.https://www.researchgate.net/publication/259440422_A_Novel_Formulation_of_Hamming_Code

Appendix:

Verilog Code, pin-planner (Quartus project file) and MATLAB Code:
Link:
https://drive.google.com/file/d/1OHwRH75VkNPEfwYhAsAcIdrtCj834NM5/view?usp=sharing